Fashion-MNIST Generative Modelling - Analysis

Week 19 Graded Mini Project · Arpit Jain · trained on mps · details and figures: notebook + results/

Model	Objective	Key result
Denoising AE (z = 32)	MSE to the clean image	PSNR 12.8 → 20.5 dB; test MSE 0.0106
VAE (z = 16)	−ELBO = BCE + KL	test −ELBO 242.0 (BCE 224.6 + KL 17.3); 16/16 dims active
cGAN (z = 100 + label)	adversarial BCE, real = 0.9	conditional accuracy 99.8% (classifier 89.3% on real); D(x) 0.69, D(G(z)) 0.21
AdaIN (StyleGAN-lite)	feature-statistics swap	mixed silhouette follows content (r = 0.63) more than style (r = 0.15)

AE vs VAE (deterministic vs probabilistic)

- The AE maps each image to a single point and only learns to reconstruct, which gives it slightly better reconstructions (MSE 0.0115 vs 0.0136) and makes it an effective denoiser (+7.7 dB PSNR).
- The VAE maps each image to a distribution and adds a KL penalty towards N(0, I). This fills the space between codes, so random samples, interpolations and traversals decode to plausible clothes. Random AE codes decode to unusable blobs.
- The price is a little blur, and factors that are entangled rather than cleanly separated.

VAE vs cGAN (likelihood vs adversarial)

- The VAE maximises a likelihood bound (BCE + KL). That covers all the data, giving diverse samples and a smooth, meaningful latent space, but per-pixel losses average details and produce blur.
- The cGAN has no reconstruction loss. A discriminator judges realism, which gives sharper edges and darker backgrounds, but it is prone to mode collapse: within-class diversity is only about 25-55% of the real data's, and there is speckle noise.
- The VAE has an encoder and a training objective you can read and log. The GAN's losses can't be read as quality; it needs visual checks plus external metrics such as the classifier accuracy and diversity scores used here.

Why labels improve control in the cGAN

- The label is given to both networks: an embedding concatenated with z in G, and a label map in D. D therefore learns "real image of this class", and G is penalised for drawing the wrong class.
- As a result, generation is steered exactly: 99.8% of samples are recognised as the requested class. In the unconditional VAE, the class can only be found by searching the latent space.
- The label splits one hard 10-mode problem into 10 easier single-class problems. This speeds up learning (class identity appears within 1-2 epochs), but it does not prevent collapse within a class.

Takeaways: what each model captures well or poorly

- **DAE:** good at global shape, clean backgrounds and noise removal. Poor at textures, logos and thin structures, and it cannot sample.
- **VAE:** good at a smooth, sample-able latent space, interpolation and coverage of all classes. Weaker at sharpness, and its factors are entangled.
- **cGAN:** good at crisp edges and exact class control. Weaker at within-class variety and fine textures (speckle), and it needs careful monitoring.
- **AdaIN:** cleanly separates content (spatial layout) from style (channel statistics), the core idea behind StyleGAN's style control. Extreme content/style mismatches distort the shape.

Practical notes

- **Learning rates:** Adam at 1e-3 (DAE) and 2e-3 (VAE) converged within a few epochs. The VAE loss flattens after about 6 epochs. The cGAN needs the slower 2e-4 with $\beta 1 = 0.5$ to keep D and G balanced.
- **Noise:** $\sigma = 0.3$ with clamping is strong enough to hide stripes. A lower σ (0.1-0.2) would preserve more texture, and a higher one makes the DAE fall back to class-average shapes.
- **Batch size:** 128 worked for all models. BatchNorm in G and D needs batches this large, and much smaller batches make GAN training noisier.

- **Stability:** one-sided label smoothing (0.9), dropout in D and a fixed-z progress grid made the GAN trainable and easy to monitor. Signs of D gradually winning call for early stopping, or for more data augmentation / a lower D learning rate.
- **Compute:** on an Apple M1 GPU (mps) the full notebook took about 23 minutes, including 18 minutes for the cGAN. On a Colab T4 it takes about 10 minutes.